
GLENCORE
Asset Criticality Analysis Standard
05

05

Asset Criticality Analysis Standard

Asset Management, Glencore Copper
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Preface

The Asset Management Framework, Policy, Manuals and Standards are intended to provide leading Asset Management practices when applied and will be continually improved for the benefit of Glencore Copper.

The real value of sharing the knowledge gained from the use of these documents will come from continued discussion of the issues and rationale behind each document rather than by reading and interpreting them as individuals.

Readers are encouraged to consider these against our values around Safety, Responsibility, Simplicity, Integrity, Openness and Entrepreneurialism and how they reflect our Purpose – to responsibly source the commodities that advance everyday life.

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1. Introduction

The Asset Management Policy, Manuals and Standards define how Glencore Copper manages the whole of life of an asset to ensure optimal value is extracted by ensuring that assets are capable of meeting business and stakeholder objectives.

Each standard provides enough detail to implement the minimum requirements at each Glencore Copper region/site. The Asset Management Framework amongst things, sets out the Glencore Copper Asset Management hierarchy of documentation of which this standard forms part of that documentation and sits at Level 3 of the hierarchy, that being corporate standards. Manual E05 Equipment Operating and Maintenance Strategy is the parent manual to this standard.

Manual E05 sets out the Level 2 process requirements of which process step 2.2 forms part of that process. This process step requires the responsible person to make a determination of Asset Criticality.

This standard provides the process by which this is to be achieved.

Figure 1-1 illustrates the Level 2 Process Flow for Element 05.

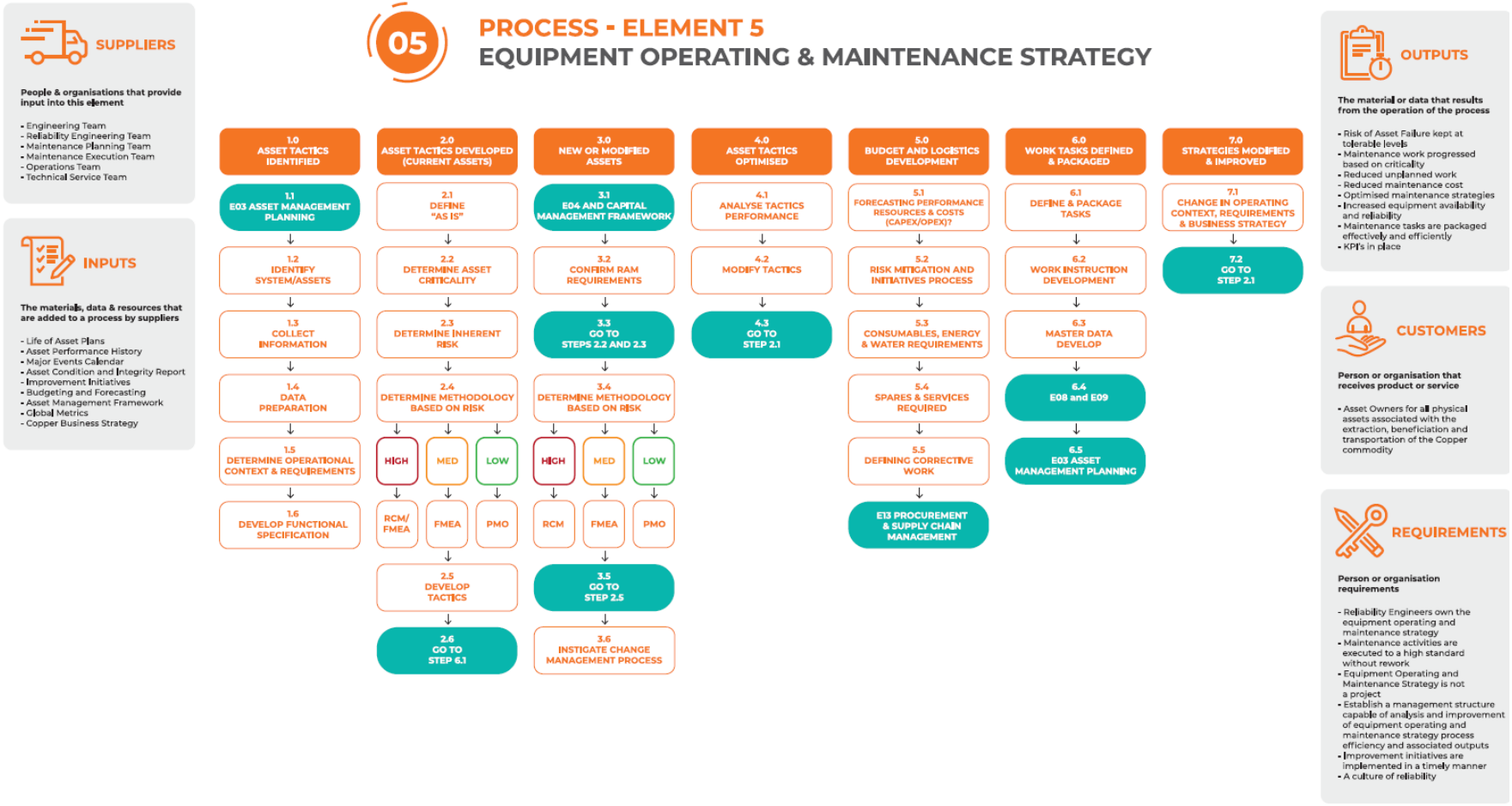


FIGURE 1-1 EQUIPMENT OPERATING AND MAINTENANCE STRATEGY PROCESS

2. Objectives

The objective of the Asset Criticality Analysis Standard is to define the process to determine the criticality of an asset according to business importance in order to guide the appropriate level of attention an asset needs across its operating life. For example, the level of effort applied in developing maintenance tactics and subsequent spares requirements, and the prioritisation of maintenance work orders.

2.1 Accountabilities

Dedicated resources must be assigned, and capabilities developed within each Operation to identify the key aspects required to ensure that all assets within each Glencore Copper region/operation are identified, criticality rank assigned, and the ranking entered into the appropriate field in the technical hierarchy of the CMMS.

Assign accountability for each or all of the following:

- System/Asset identification/boundaries.
- Criticality Analysis down to Maintainable Item Level
- Populating Criticality Rank into CMMS
- Reviewing the most critical items, that being A ranked items on the ABC scale, annually to ensure that the Criticality Rank remain relevant.

2.2 Scope

The Glencore Copper Corporate Risk Management Framework provides a methodology for defining the risks related to asset failures. It utilises a well-defined consequence matrix to determine the impacts of asset failures regarding safety, environment, production, failure to meet regulatory and legal requirements and reputation. It also utilises a well-defined likelihood factor that when combined with the consequence factor provides a risk ranking. This framework has been adopted for the development of Asset Criticality and modified to account for local risk impact. This is called the Glencore Copper Asset Criticality Risk Matrix. Appendix A provides a copy of the Matrix.

The Criticality Analysis uses the site's functional location and equipment hierarchy as illustrated in Figure 2-1 below starting at the functional location sub-process level (e.g., Primary Grinding or Loader Fleet). If the functional location sub-process level is deemed Criticality "A", the criticality analysis moves down into equipment level 1, referred to as "Maintainable Unit" (e.g., SAG Mill or Loader LD201). The same is then done for equipment level 2, referred to as "Maintainable Function" (e.g., Lube System or Engine), and then for equipment level 3, referred to as "Sub Maintainable Function" (e.g., Pump or Fuel System).

Further, critical spares can be assessed at this level, only if the maintainable function has been assessed as critical. This is managed through the Glencore Copper Critical Spares Standard.

This top-down approach provides significant time benefit in implementing the process as lower-level items can be discounted from the process if the upper-level items have been assessed as non-critical, that being B and C ranked items on the ABC indicator scale.

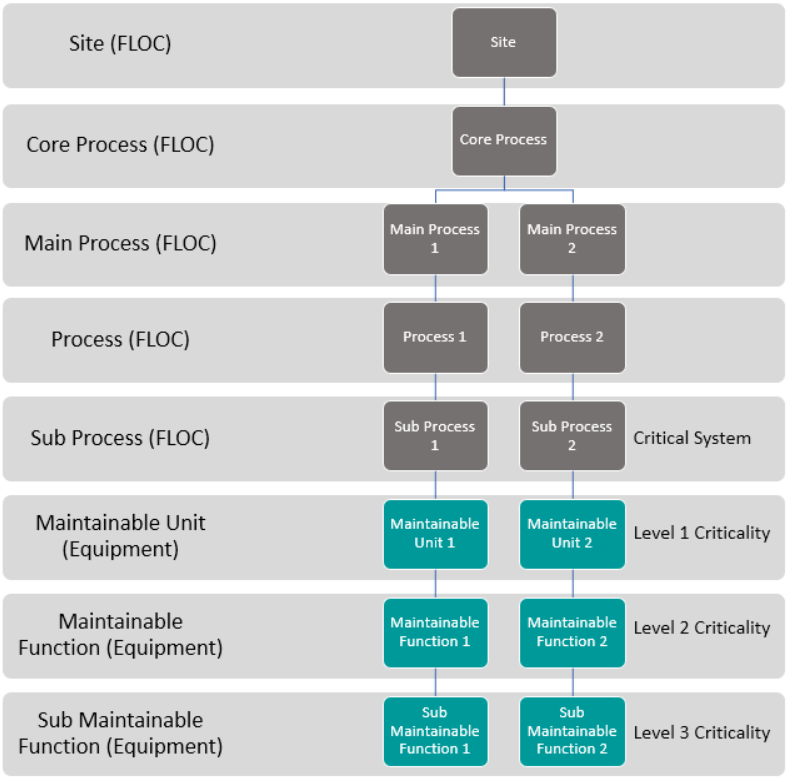


FIGURE 2-1 TOP DOWN FOUR LEVEL ANALYSIS

In order to understand the true impact of functional failures at site level some adjustment of the Glencore Copper Risk Matrix has been undertaken. The financial impact in particular as detailed in the Corporate Matrix would in effect deem all items as non-critical if it were used in the analysis of items locally on site.

Therefore, the financial impact in the risk matrix has been adjusted to meet site operational consequence requirements and the Glencore Copper Asset Criticality Matrix has been developed to enable criticality to be asset based on local impacts.

Note: At Sub Process level, the term Sub Maintainable Function is used to refer to individual assets. References to "asset" in this procedure will therefore be considered to include the term "sub maintainable function". Also, Sub Process, Maintainable Unit, Maintainable Function and Sub Maintainable Function Criticality will be collectively referred to as Criticality.

Criticality Analysis must be utilised within Operations for the development and continual improvement for the operation and maintenance of Glencore Copper equipment.

Inherent risk as determined in the criticality analysis process is to be converted to the Glencore Copper 3 level ABC Criticality criteria. Residual risk will be determined through the RCM/FMEA/PMO process, as detailed in the Glencore Copper Maintenance Tactic Development Standard.

The scope of Criticality Analysis includes:

Determination of the most critical assets within the Glencore Copper portfolio to enable:

- Focus on those assets that could have the biggest impact to the Glencore Copper business in terms of Health, Safety, Environment, Financial, Image, Reputation, Legal and Compliance.

- Input into the development of Equipment Operating and Maintenance Strategies as prescribed in Element E05 of the Asset Management Framework and the Glencore Copper Maintenance Tactics Development standard
- Input into the identification of critical spares.
- Input into the sustaining Capital process.
- Input into the priority for executing work.

3. Definitions

Asset – an item, thing or entity that has potential or actual value to an organisation.

Asset Criticality Ranking (ACR)– a process that enables the ranking of systems assets in order of business importance to ensure that effort applied is directionally proportionate to the criticality and impact to the business. Criticality is relevant for the business decision making process to manage business risk and value, Criticality is typically used:

- As an input to determine the overall priority for performing a maintenance task (sometimes combined with a "Work Order Priority" entered against the specific task to give an overall priority for the task)
- To determine, at a high level, the type of risk mitigation strategy to be applied to the equipment (e.g., do condition monitoring and defect elimination on high criticality items)
- As an input into determining the optimum spare parts holdings required for the equipment item
- To provide input into the capital program so that "high criticality" equipment is given a higher priority for upgrade or replacement.
- To guide reliability engineers so that they focus their reliability improvement efforts on the most "critical" equipment.

The criticality analysis process does not include the analysis of items that have been flagged for deletion in the CMMS and of failure modes this is managed through the Glencore Copper Maintenance Tactics Development Standard.

Critical Asset - asset having potential to significantly impact on the achievement of the organisation's objectives". Those assets that have a criticality ranking of A from the Glencore Copper SAP Master Data Governance Standard.

Critical System – a maintainable unit comprising of maintainable and sub maintainable functions (critical equipment) that if unavailable would have a significant risk impact to Glencore Copper.

Criticality Analysis - is a process by which assets are assigned a criticality rating based on their potential risk.

Inherent Risk - is typically defined as the level of risk in place in order to achieve an organisations objective and before actions are taken to alter the risk's impact or likelihood.

Maintainable Function – an asset that produces part of the overall output. e.g., haul truck engine, a pump motor

Maintainable Unit– a group of assets that produces an output from an input. e.g., a haul truck, a pump (pump, motor, coupling, instruments etc.)

Residual Risk - is the remaining level of risk following the development and implementation of the organisation response.

Significant Impact – an event that could potentially cause a risk ranking of 17 (High) or more on the Glencore Copper Asset Criticality matrix.

Sub Process - The means by which resources are transformed from inputs to create useful goods and services as outputs.

4. Approach

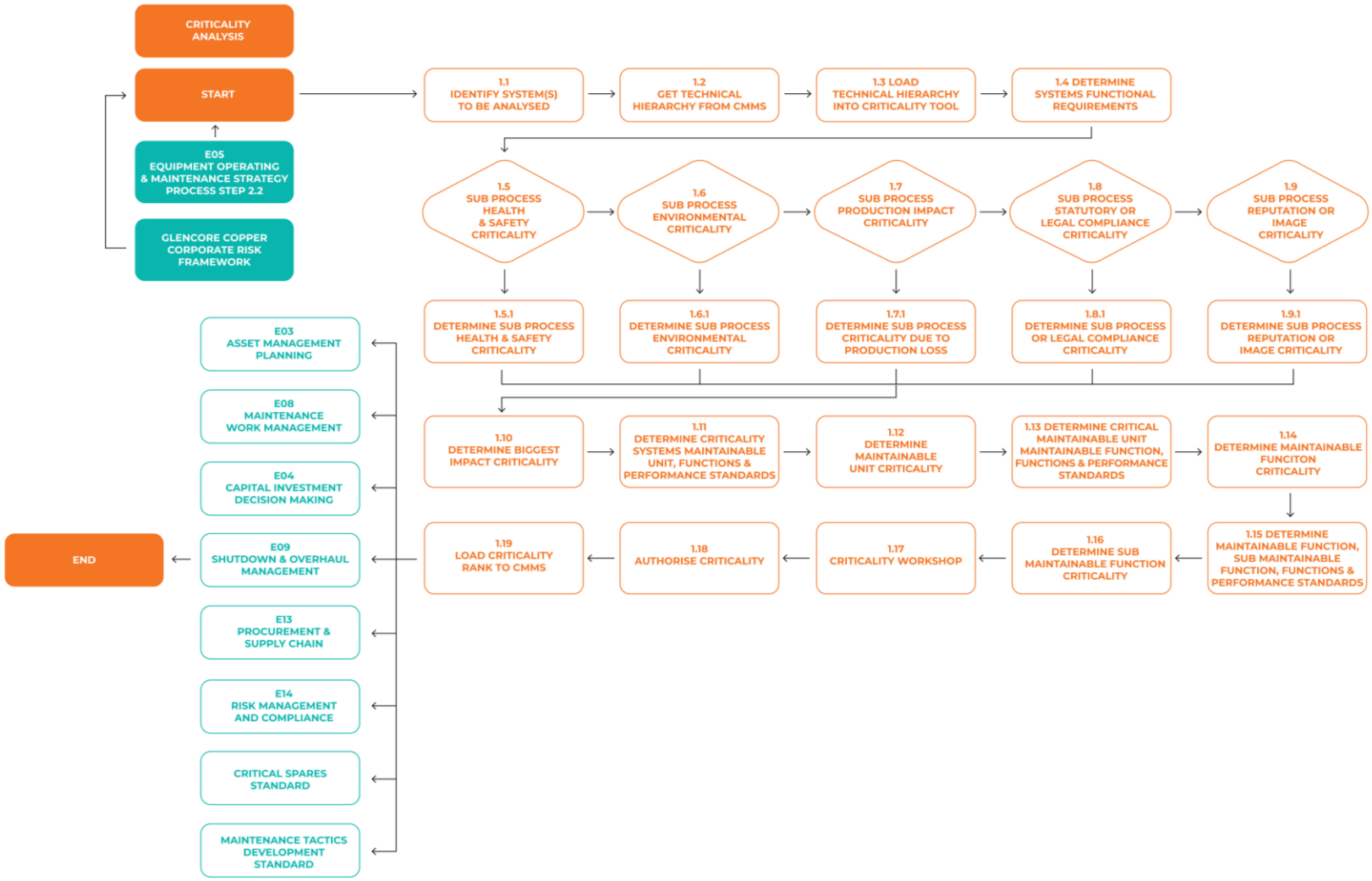
The approach for Criticality Analysis is described through:

- Process Flow - pictorially illustrates the key Criticality Analysis activities, refer to Figure 4-1.
- Process Description – defines in more detail the key Criticality Analysis activities, refer to Section 4.2.
- Process RACI Matrix - indicates for each activity, the functions who are Responsible, Accountable, who are Consulted prior to final decision/action, and those who are Informed of any outcomes and actions, refer to Section 4.3

Asset Criticality Analysis Standard

4.1 Process Flow

FIGURE 4-1 ASSET CRITICALITY ANALYSIS PROCESS FLOW



4.2 Process Description

TABLE 4-1 ASSET CRITICALITY ANALYSIS PROCESS DESCRIPTION

PROCESS STEP	DESCRIPTION
1.1 Identify Sub Process/es to be Analysed	Sub Process/s shall be identified that do not have criticality ranking assigned, or that the current ranking has not been re-assessed during the previous 12 months. Once all assets have been assigned a criticality in the CMMS then only those assets of Criticality rank A require annual review as per Section 7 of this manual.
1.2 Get Technical Hierarchy from CMMS	Download the Technical Hierarchy from the CMMS for those systems to be analysed.
1.3 Load Data into Criticality Tool	The technical hierarchy is to be loaded into the Criticality tool, together with all other information deemed necessary for the analysis. This could include the following: • Failure History • Historical Cost • Performance criteria (Downtime, MTBF, Availability, Reliability Etc.)
1.4 Determine Sub Process Functional and Performance Standards	In order to understand when a sub process has begun to or has entered functional failure then knowledge of the functional and performance parameters is required for that sub process. Both Primary and Secondary functions should be developed. For example, a conveying sub process Primary function is to transport product from A to B at a feed rate of x tonnes/hour and a secondary function is for the product to be contained within the system. By establishing the function and performance parameters then the analyst can easily decide which sub processes are important from a functional perspective.
1.5 Sub Process Health and Safety Criticality	The health and safety impact of the system according to Glencore Copper Criticality Risk Matrix.
1.5.1 Determine Sub System Health and Safety Criticality	From the Glencore Copper Criticality Analysis Risk Matrix determine the maximum risk value. To be able to determine system criticality consideration must be given to the following: **Asset Failure** An asset is considered to have failed when it can no longer provide the function it was intended to deliver. The reason for this could be due to a physical asset failure (it broke), due to other external influences like flood inundation, lightning strike etc. Assets are also considered to have failed due to a reduction in delivery capacity (performance) also referred to as Functional Failure. **Consequence of failure**

At Sub Process Level the consequence of the failure of an asset is the effect that the failure of that asset will have on health, safety, the environment, production, statutory or legal compliance and reputation or image on the business. The Glencore Copper Consequence Descriptor Matrix (refer Appendix A, Table 2) uses various consequence categories to better understand and define the overall consequence of a failure event.
When the consequence of failure for Sub Processes are assessed, each consequence category is considered separately and rated in accordance with the descriptors in the Consequence Descriptor Matrix on a 5-level scale that ranges between 1 (*Negligible)* and 5 (*Catastrophic)*. This matrix makes it possible to compare the consequence of failure of different sub processes on the same basis and can therefore be applied across Glencore Copper's entire asset base.
The aim of assessing the consequence of failure at Sub Process Level is to determine what the relative impact of an asset failing is on the Sub Process's operation and safety and thus identify the assets that are critical for the safe and satisfactory performance of the System.

Likelihood of Failure
The likelihood of an asset failing to deliver its intended function could be affected by various factors.
The likelihood of asset failure is ranked on a scale that aligns with Glencore Copper's Criticality Analysis Risk Process (refer Appendix A, Table 3). The scale ranges between E (*Rare*) and A (*Almost Certain*).

Criticality Evaluation
The consequence and the likelihood of Sub Process failure are used to define the failure risk in accordance with Glencore Copper's Criticality Analysis Risk Matrix (refer Appendix A, Table 4). These risks represent the impact to the business and are used to determine business decisions.

When the consequence of sub process failure is determined in accordance with the Corporate Consequence Descriptor Matrix, the highest consequence rating in any of the categories is used together with the likelihood of the failure event to define the event risk. It is proposed that the consequence ratings for the different categories are not added for a combined event consequence rating.

For Sub Processes that have a ranking less than 17 (High) no further analysis is required. Items greater than 17 (High) should have a second level of analysis conducted at the Productive Unit Level. Risk scores are to be converted to ABC ranking for inclusion in the CMMS as per the table below:

Risk Ranking	ABC Ranking	Guiding Principals (% asset base +/- 10%)
17- 25 (High)	A	20
7 – 16 (Medium)	B	30
1 – 6 (Low)	C	50

Note: Guiding principals provide an indicative % of the asset base that could be expected as a result of the analysis. Actual results may differ.

1.6 Sub Process Environmental Criticality	The environmental impact of the sub process according to Glencore Copper Criticality Risk Matrix.
1.6.1 Determine Sub Process Environmental Protection Criticality	From Glencore Copper Criticality Analysis Risk Matrix determine the maximum risk value. Refer to Step 1.5.1
1.7 Sub Process Production Impact Criticality	The production impact of the Sub Process according to Glencore Copper Criticality Risk Matrix.
1.7.1 Determine Sub Process Criticality due to Lost Production?	From the Glencore Copper Criticality Analysis Risk Matrix determine the maximum risk value. Refer to Step 1.5.1
1.8 Sub Process Statutory or Legal Compliance Criticality	The legal and statutory compliance impact of the sub process according to Glencore Copper Criticality Risk Matrix.
1.8.1 Determine Sub Process Statutory or Legal Compliance Criticality	From the Glencore Copper Criticality Analysis Risk Matrix determine the maximum risk value. Refer to Step 1.5.1
1.9 Sub Process Reputation or Image Criticality	The reputation or image impact of the sub process according to Glencore Copper Criticality Risk Matrix.
1.9.1 Determine Reputation or Image Criticality	From the Glencore Copper Criticality Analysis Risk Matrix determine the maximum risk value. Refer to Step 1.5.1
1.10 Determine Biggest Impact Criticality	From the assessment done at process steps 1.5 through 1.9 determine which has the highest risk impact, this impact is to then be further assessed for Productive Assets within the Critical System.
1.11 Determine Critical Sub System Maintainable Unit Functional and Performance Standards	In order to understand when a maintainable unit has begun to or has entered functional failure then knowledge of the functional and performance parameters is required for that maintainable unit. Both Primary and Secondary functions should be developed. For example, a pump's Primary function is to transport product from A to B at a pressure of X KPa and feed rate of x litres/minute and a secondary function is for the product to be contained within the system. By establishing the function and performance parameters then the analyst can easily decide which system are important from a functional perspective.
1.12 Determine Maintainable Unit Criticality	Using the highest criticality ranking for the critical system determine the criticality for all of the maintainable units in that maintainable unit. Those maintainable units with a ranking of 17 (High) or more on the Glencore Copper Criticality Analysis Risk Matrix shall have further critical analysis conducted at the maintainable function level. No further analysis is required for all non-critical maintainable units.
1.13 Determine Critical Maintainable Unit, Maintainable Function Functions and Performance Standards	A third level of Functions and Performance analysis is to be conducted for all Maintainable functions of critical maintainable units.

1.14 Determine Maintainable Function Criticality	Similar to determining Maintainable Unit criticality Maintainable Function criticality needs to be determined in the same manner.
1.15 Determine Critical Maintainable Function, Sub Maintainable Function Functions and Performance Standards	Determine the functions and performance standards of the critical Maintainable Functions, Sub Maintainable Functions
1.16 Determine Sub Maintainable Function Criticality	Similar to determining Maintainable Function criticality Sub Maintainable Function criticality needs to be determined in the same manner.
1.17 Criticality Workshop	The Criticality Lead is to form a multi-functional team consisting of: • Maintenance • Planning • Operations • Engineering • Reliability • Process A workshop or workshops shall be conducted with the intent of the workshop is for team members to challenge/validate the criticality assessment results. Generally, there would only be a need to challenge/validate the lowest level criticality results (maintainable assets) and higher-level asset adjusted as necessary.
1.18 Authorise Criticality	The resultant workshop criticality results shall be reviewed and authorised by the accountable person. A criticality report is to be developed detailing results.
1.19 Load Criticality Rank to CMMS	The risk score from the Glencore Copper Criticality Analysis Risk Matrix is to be converted into a format that can be uploaded and used in the CMMS. For example, in SAP the ABC indicator on the Functional Location provides functionality to assign the criticality to the System, Productive and Maintainable assets. This conversion should align to the risk level in the Glencore Copper Criticality Analysis Risk Matrix of Low, Medium and High Risk, which also aligns to the requirement of the risk assessment required as part of the implementation of Element 05 Equipment Operating and Maintenance Strategy.

4.2.1 Criticality Example

Refer to Appendix B for examples of criticality assessment.

4.3 Process RACI Description

The RACI matrix below in Table 4-2 addresses the high-level functional elements of the Criticality Analysis process.

Site roles may differ to the roles identified in this RACI, these roles however should be used as a guide for the development of a site-specific RACI, while maintaining the intent and content of this Standard.

Note: The HSE and Commercial Departments do not appear in this high-level RACI, although their involvement will be required from time to time dependent on the nature of the variance.

The codes used in the RACI analysis are defined as follows:

TABLE 4-2 PROCESS RACI

R	Responsible	The "doer" – the individual(s) who will perform an activity or directs others to perform the activity.
A	Accountable	"The buck stops here" –the individual who is ultimately "owns" the activity – there should only be one person
C	Consulted	person that needs to feedback and contribute to the activity
I	Inform	The individual(s) who needs to be informed after a decision or action is taken if relevant. (via 1-way communication)

Process Step \ Functional Role	Reliability Management	Maintenance Management	Master Data Analysis	Maintenance Execution	Reliability Engineering	Maintenance Planning	Technical Services
1.1 Identify Sub Process/es to be Analysed	A	C		C	R	C	C
1.2 Get Technical Hierarchy from CMMS	A				C	R	
1.3 Load Technical Hierarchy to Criticality Tool	A				R		
1.4 Determine Sub Process Functional and Performance Standards	A		C	C	R	C	C
1.5 Sub Process Health and Safety Criticality							
1.5.1 Determine Sub Process Health and Safety Criticality	A	C	I	C	R	C	C
1.6 Sub Process Environmental Criticality							
1.6.1 Determine Sub Process Environmental Criticality	A	C	I	C	R	C	C
1.7 Sub Process Production Impact Criticality							
1.7.1 Determine Sup Process Production Impact Criticality	A	C	I	C	R	C	C
1.8 Sub Process Statutory or Legal Compliance Criticality							
1.8.1 Determine Sub Process Production Impact Criticality	A	C	I	C	R	C	C
1.9 Sub Process Reputation or Image Criticality							
1.9.1 Determine Sub Process Reputation or Image Criticality	A	C	I	C	R	C	C
1.10 Determine Biggest Impact Criticality	A				R		

Process Step \ Functional Role	Reliability Management	Maintenance Management	Master Data Analysis	Maintenance Execution	Reliability Engineering	Maintenance Planning	Technical Services
1.11 Determine Critical Sub System, Maintainable Unit Functions and Performance Standards	A		C	C	R	C	C
1.12 Determine Maintainable Unit Criticality	A	C	I	C	R	C	C
1.13 Determine Critical Maintainable Unit Maintainable Function, Functions and Performance Standards	A		C	C	R	C	C
1.14 Determine Maintainable Function Criticality	A	C	I	C	R	C	C
1.15 Determine Maintainable Function, Sub Maintainable Function, Functions and Performance Standards	A		C	C	R	C	C
1.16 Determine Sub Maintainable Function Criticality	A	C	I	C	R	C	C
1.17 Criticality Workshop	A	I	C	C	R	C	C
1.18 Authorise Criticality	A	I	I	I	R	I	I
1.19 Load Criticality Rank to CMMS	A	I	R	I	C	R	I

TABLE 4-2 FUNCTIONAL DESCRIPTIONS

Function	Description
Reliability Management	• Ensures that the Criticality Analysis standard is implemented across site. • Ensures that Criticality ranking on all assets are authorised. • Ensures that Criticality ranks are applied to all assets registered in the CMMS. • Ensures that this standard is reviewed annually. • Ensures that A critical assets are reviewed annually.
Maintenance Management	• Provides input into the implementation of the Criticality Analysis standard. • Provides advise on the criticality of Sub Processes, Maintainable Units, Maintainable Functions and Sub Maintainable Functions.
Maintenance Execution	• Provides input into the Criticality Analysis standard is implemented across site. • Provides input into the Criticality ranking on all assets are authorised. • Provides input regarding Criticality ranks being applied to all assets registered in the CMMS. • Provides input into this standard being reviewed annually. • Provides input regarding A critical asset being reviewed annually.
Reliability Engineering	• Responsible for the Criticality Analysis standard being implemented across site.

	• Responsible for Criticality ranking on all assets being authorised. • Responsible for Criticality ranks being applied to all assets registered in the CMMS. • Responsible for this standard being reviewed annually. • Responsible for A critical assets being reviewed annually.
Maintenance Planning	• Provides input into the Criticality Analysis standard is implemented across site. • Provides input into the Criticality ranking on all assets are authorised. • Provides input regarding Criticality ranks being applied to all assets registered in the CMMS. • Provides input into this standard being reviewed annually. • Provides input regarding A critical asset being reviewed annually. • Responsible for loading master data to the CMMS.
Master Data Analysis	• Provides input into the Criticality Analysis standard is implemented across site. • Provides input into the Criticality ranking on all assets are authorised. • Provides input regarding Criticality ranks being applied to all assets registered in the CMMS. • Provides input into this standard being reviewed annually. • Provides input regarding A critical asset being reviewed annually. • Responsible for loading master data to the CMMS.
Technical Services – Metallurgists, Process Engineers and Mining	• Provides input into the Criticality Analysis standard is implemented across site. • Provides input into the Criticality ranking on all assets are authorised. • Provides input regarding Criticality ranks being applied to all assets registered in the CMMS. • Provides input into this standard being reviewed annually. • Provides input regarding A critical asset being reviewed annually. • Responsible for loading master data to the CMMS.

5. Deployment

The deployment of the Criticality Analysis approach across all business areas is through:

- Awareness presentation
- Position description alignment
- Competency assessment
- Training, coaching, and mentoring

5.1 Awareness Presentation

This standard is supported by a slide presentation and video to help facilitate communication and to raise awareness on how it contributes to the Equipment Operating and Maintenance Strategy and business success.

5.2 Position Description Alignment

To effectively deploy this standard, the essential process steps and activities need to be diligently mapped and assigned to approved positions within Glencore Copper.

The RACI matrix assists in the visualisation that everyone is working towards the same goal. It clarifies, not only the individual's responsibilities, but also that of others. It has proven to be an effective tool to quickly understand and close the gaps between current and desired practices and business process activities.

Position descriptions including KPIs need to be aligned to the RACI matrix and measures of success. This is the basis for ongoing performance reviews and the creation and execution of individual development plans.

5.3 Competency Assessment

This standard is supported by an Asset Management competencies assessment process and each site is responsible for assessing Asset Management competencies:

- Guide recruitment, selection, development, and deployment of people in asset management roles across the company
- Provide employees responsible for managing assets with feedback, development, coaching and recognition for their performance.
- Map, build and manage the asset management talent pool across the company.
- Identify strengths and development needs in asset management practitioners.

5.4 Training, Coaching and Mentoring

Coaching and mentoring is another way to assist employees improve their understanding of their role and to improve performance.

Coaching is generally about improving performance in the short term where mentoring is about achieving high performance and career aspirations over a longer term.

Coaching or mentoring requires addressing that a person is not currently up to the required level to perform their role as effectively and efficiently as they could. The build-up of skills to perform efficiently and effectively requires a large investment of time and cooperation.

Coaches and Mentors need to be carefully selected and must be committed for the long term to effect maximum success.

6. Improvement

The process to review and continually improve the approach and deployment for Criticality Analysis is scheduled annually or when there is a sudden change in the Business and Asset Management objectives. As the organization matures a longer review period could be considered.

7. Criticality Review

Criticality Rankings should be reviewed annually for Critical A items or when significant changes occur in the business that could prompt a change in criticality ranking, for example the introduction of redundancy in Processes and Sub Processes. All other items should be reviewed as operational context changes.

8. References

The following documents provide detailed support material that should be read in conjunction with this Standard to provide a comprehensive understanding of Criticality Analysis.

DOCUMENT TITLE	DESCRIPTION
Asset Management Framework	This document defines the overall direction of asset management across Glencore Copper.
Equipment Operating and Maintenance Strategy Development Manual Element 05	This document defines the process for the development and implementation of Equipment Operating and Maintenance Strategies within Glencore Copper
Glencore Copper Criticality Analysis Risk Matrix	The multi-criteria risk matrix that is used to determine corporate risk through consequence and likelihood evaluation of potential failure events.

9. Revision History

VERSION NO.	REASON FOR REVISION	DATE
0.1	New Procedure to support Asset Management Manual	07/07/2021
0.2	Gather Documents and Create Draft Procedure	22/07/2021
0.3	Review and Signoff 1st Draft Procedure with Sites	11/08/2021
0.4	Reformat and Issue for Final Review	14/09/2021
1.0	Update Site Signoff and Publish	

10. Appendix A – Glencore Copper Asset Criticality Risk Matrix

GLENCORE CORPORATE RISK MATRIX

CONSEQUENCE [potential foreseeable outcome of the event]

	Health & Safety	Environment	Financial Impact	Image & Reputation / Community	Legal & Compliance
5 Catastrophic	• Multiple fatalities (5 or more fatalities in a single incident) • Multiple cases (5 or more) of Permanent Damage Injuries or Diseases that result in permanent disabilities in a single incident	• Unconfined and widespread Environmental damage or effect (permanent, >10 years) • Requires major remediation	• Revenue loss > 50M U$ (3 months of operation or higher)	• Loss of multiple major customers or large proportion of sales contracts • Sustained campaign by one or more international NGOs resulting in physical impact on the assets or loss of ability to operate • Security incident resulting in multiple fatalities or major equipment damage • Formal expression of significant dissatisfaction by government • Grievance from internal or external stakeholder alleging human rights violation resulting in multiple fatalities	• Major litigation / prosecution at Glencore corporate level • Nationalisation / loss of licence to operate
4 Major	• Single incident resulting in: Less than 5 Fatalities • Permanent Damage Injury or Disease that results in a permanent disability- less than 5 cases in a single incident	• Long-term (2 to 10 years) impact • Requires significant remediation	• Revenue loss > 15M and < 50M U$ (2 weeks to 3 months of operation)	• Security/ stakeholder incident resulting in single loss of life or equipment damage • Grievance from internal or external stakeholder alleging human rights violation resulting in single fatality or serious injuries • Topic of broad societal concern and criticism • Negative media coverage at international level resulting in a Corporate statement within 24 hours • Investigation from government and/ or international (or high-profile) NGOs • Complaints from multiple "final" customers • Loss of major customer • Negative impact on share price	• Major litigation / prosecution at Department level
3 Moderate	• Lost Time Injury (LTI) • Lost Time Disease (LTD) • Permanent Disabling Injury (PDI) • Permanent Disabling Disease (PDD) • Single incident that results in multiple medical treatments	• Medium-term (<2 years) impact (typically within a year) • Requires moderate remediation	• Revenue loss > 5M and < 15M U$ (5 days to 2 weeks of operation)	• Negative media coverage at national level over more than one day • Complaint from a "final" customer • Off-spec product • Local Stakeholder action resulting in national societal scrutiny	• Major litigation / prosecution at Operation level
2 Minor	• Medical Treatment Injury (MTI) • Medical Treatment Disease (MTD) • Restricted Work Injury (RWI) • Restricted Work Disease (RWD)	• Near source • Short-term impact (typically <week) • Requires minor remediation	• Revenue loss > 1M and < 5M U$ (1 to 5 days of operation)	• Negative local/ regional media coverage • Complaint received from an internal or external stakeholder	• Regulation breaches resulting in fine or litigation
1 Negligible	• First Aid Injury (FAI) or illness (not considered disease or disorder)	• Near source and confined • No lasting environmental damage or effect (typically <day) • Requires minor or no remediation	• Revenue loss < 1M U$ (equivalent to 1 day of operation)	• Negligible media interest	• Regulation breaches without fine or litigation

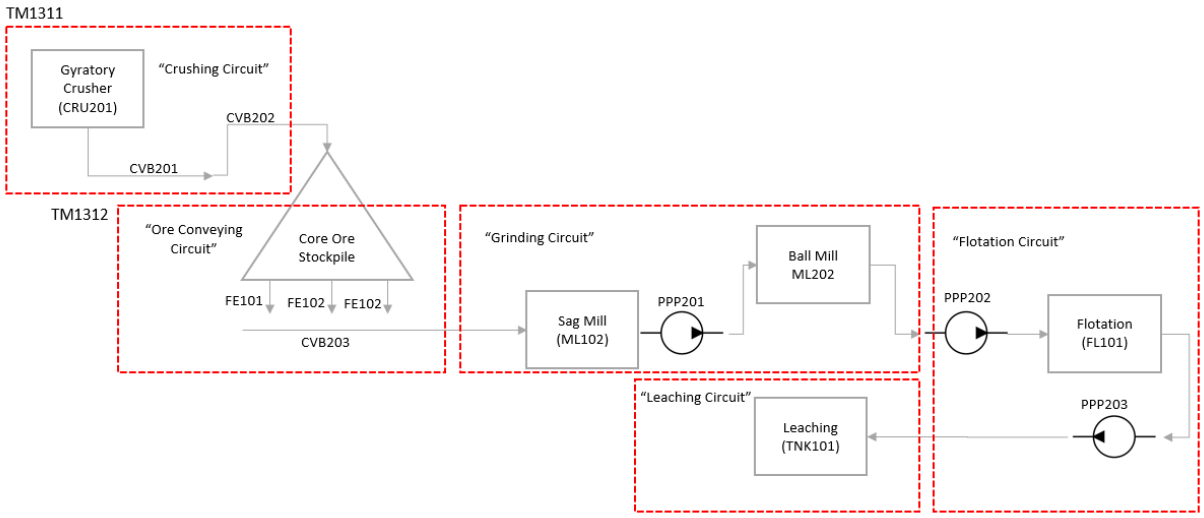
LIKELIHOOD [of the event occurring with that consequence]

Basis of Rating	E - Rare	D - Unlikely	C - Possible	B - Likely	A – Almost Certain
LIFETIME OR PROJECT OR TRIAL OR FIXED TIME PERIOD OR NEW PROCESS / PLANT / R&D	Unlikely to occur during a lifetime OR Very unlikely to occur OR No known occurrences in broader worldwide industry	Could occur about once during a lifetime OR More likely NOT to occur than to occur OR Has occurred at least once in broader worldwide industry	Could occur more than once during a lifetime OR As likely to occur as not to occur OR Has occurred at least once in the mining / commodities trading industries	May occur about once per year OR More likely to occur than not occur OR Has occurred at least once within Glencore	May occur several times per year OR Expected to occur OR Has occurred several times within Glencore
	15 (M)	19 (H)	22 (H)	24 (H)	25 (H)
	10 (M)	14 (M)	18 (H)	21 (H)	23 (H)
	6 (L)	9 (M)	13 (M)	17 (H)	20 (H)
	3 (L)	5 (L)	8 (M)	12 (M)	16 (M)
	1 (L)	2 (L)	4 (L)	7 (M)	11 (M)

Consequence Category	Consequence Type	Ownership	Action
Cat. 5	Catastrophic Hazard	Department / Functional / Operational / Asset Leadership	• Quantitative or semi-quantitative risk assessment required. • Capital expenditure will be justified to achieve ALARP ("As Low As Reasonably Practicable"). • Catastrophic Hazard Management Plans (CHMP) must be implemented where practical, Crisis Management Plans (CMP) tested and Catastrophic Event Recovery Plans (CERP) developed.
Cat. 4 (Health & Safety consequence)	Fatal Hazard	Department / Functional / Operational / Asset Leadership	• Glencore SafeWork Fatal Hazard Protocols or appropriate management plans must be applied. • Capital expenditure will be justified to achieve ALARP.
Risk Rank	**Risk Rating**	**Ownership**	**Action**
17 to 25	High Risk	Department / Functional / Operational / Asset Leadership	• Install additional HARD and SOFT controls to achieve ALARP. • Capital expenditure will be justified to achieve ALARP.
7 to 16	Medium Risk	Operational / Asset Leadership	• Install additional HARD and SOFT controls if necessary to achieve ALARP. • Capital expenditure may be justified.
1 to 6	Low Risk	Operational / Asset Leadership	• Install additional controls if necessary to achieve ALARP. • Capital expenditure is not usually justified.

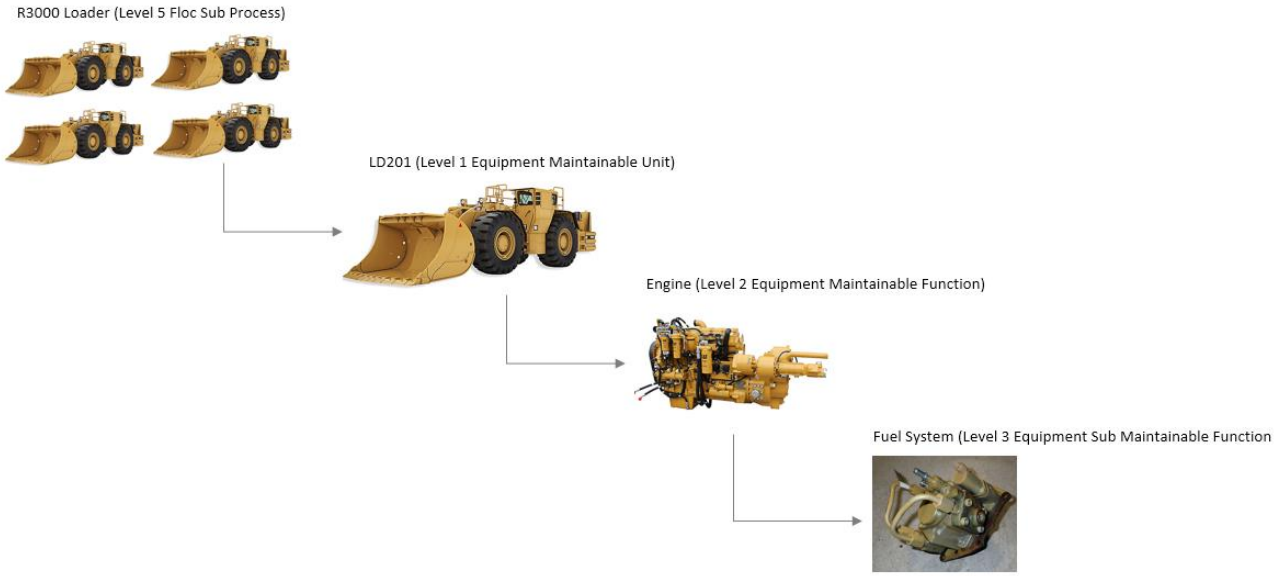
11. Appendix B – Examples Criticality Analysis

Below is an example of a Fixed Plant Criticality Analysis



			Criticality Factors (Consequence x Likelihood)						
Location Level	**Asset Number**	**Asset Description**	**Health and Safety**	**Environment**	**Financial**	**Image and Reputation**	**Legal and Compliance**	**Criticality Ranking**	**ABC Indicator**
Sub Process	TM1311	Crushing Circuit	1	1	20	1	1	20	A
Maintainable Unit	TM1311-CRU201	Gyratory Crusher	1	1	20	1	1	20	A
Maintainable Function	TM1311-CRU201 – MTR201	Motor, Gyratory Crusher	1	1	20	1	1	20	A
Sub Process	TM1312	Ore Conveying Circuit	1	1	15	1	1	15	B
Maintainable Unit	TM1312-CVB203	Conveyor 203	1	1	15	1	1	15	B
Maintainable Function	TM1312-CVB203-GRB203	Gearbox, Conveyor 203	1	1	15	1	1	15	B
Sub Maintainable Function	TM1312-CVB203-GRB203-CPL203	Coupling, Conveyor 203	1	1	15	1	1	15	B

Below is an example of a Mobile Plant Criticality Analysis



			Criticality Factors (Consequence x Likelihood)						
Location Level	**Asset Number**	**Asset Description**	**Health and Safety**	**Environment**	**Financial**	**Image and Reputation**	**Legal and Compliance**	**Criticality Ranking**	**ABC Indicator**
Sub Process	R3000	R3000 Loader	1	1	20	1	1	20	A
Maintainable Unit	LD201	Loader 201	1	1	20	1	1	20	A
Maintainable Function	Engine	Engine, Loader 201	1	1	20	1	1	20	A
Sub Maintainable Function	Fuel System	Fuel System, Engine, Loader 201	1	1	20	1	1	20	A

12. Appendix C – Stakeholder Sign-Off

Stakeholder Name	Site or Department	Contact	Stakeholder Signoff
Rohan Reisener	Glencore Copper Asset Management	Rohan.Reisener@glencore.com.au	
Dino Moll	Glencore Copper Asset Management	Dino.Moll@glencore.com.au	
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Olivier Cote	Horne	Olivier.Cote@glencore.ca	
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Claudia Bernatchez	CCR	Claudia.Bernatchez@glencore.ca	
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Ghislain Bisson	SAP PM Global Business Process Owner	Ghislain.Bisson@glencore.ca	
Akeem Araoye	Glencore Copper Asset Management – North America	Akeem.Araoye@glencore.ca	
Yuri Yurac	Glencore Copper Asset Management – South America	Yuri.Yurac@glencore.cl	
Anthony Green	Glencore Copper Asset Management – Australasia	Anthony.Green@glencore.com.au	
Lawrence Hlatshwayo	Glencore Copper Asset Management – Africa	Lawrence.Hlatshwayo@glencore.co.za	


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Asset Criticality Analysis Standard

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